
CSE322

Equivalence of DFA and

NDFA

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Acceptability

- NFAs and DFAs are equivalent in that if a language is recognized by an NFA, it is also recognized by a DFA and vice versa.
- Note: **Every NDFA is DFA but vice versa is not true**

Equivalence of DFA and N DFA

We naturally try to find the relation between DFA and N DFA. Intuitively we now feel that:

- (i) A DFA can simulate the behaviour of N DFA by increasing the number of states. (In other words, a DFA $(Q, \Sigma, \delta, q_0, F)$ can be viewed as an N DFA $(Q, \Sigma, \delta', q_0, F)$ by defining $\delta'(q, a) = \{\delta(q, a)\}$.)
- (ii) Any N DFA is a more general machine without being more powerful.

Conversion from NFA to DFA may provide the following results,

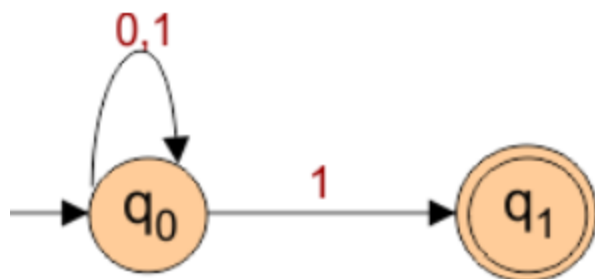
- The number of states in the resulting DFA may differ from that of the original NFA.
- The DFA can have a maximum of 2^n states where “n” is the number of states in the NFA.

Converting NFA to DFA Algorithm

- **Step 01:** Draw an NFA graph (**Mostly Given**).
- **Step 02:** Draw the NFA transition table from the NFA diagram.
- **Step 03:** Convert the NFA transition table to the DFA transition Table.
 - **Step 3.1:** Select the first row of the NFA table to begin constructing the DFA. This row will now serve as the initial rows of the DFA.
 - **Step 3.2:** If any state appears in the input column of the DFA but is not found in the States column of the DFA, it is considered a **New State**. Simply copy this new state to the states column and add its transitions in the DFA table. The transitions for the New State in the DFA table will follow the NFA transition table. Repeat this procedure until no new states are found in the input columns.
 - **Step 3.3:** If “x” was the final state in the NFA Transition Table, then all those states will be the final states in the DFA Transition Table where “x” exists.
- **Step 04:** Convert the DFA table to the DFA Diagram.

Example 1

Solution: Draw the NFA transition table from diagram



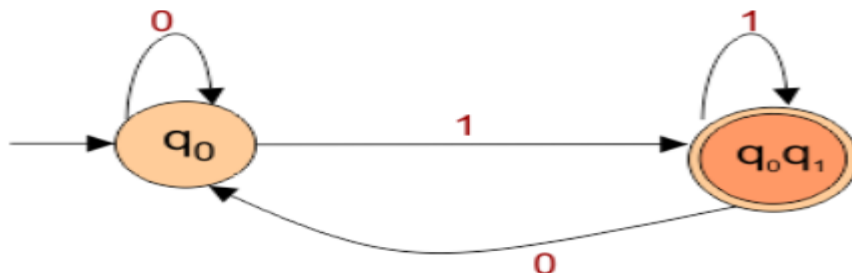
States	0	1
→ q0	q0	q0q1
* q1	-	-

Select the first row of the NFA transition table, which will become the first row of the DFA transition table given below.

States	0	1
q0	q0	q0q1

State “**q0q1**” is a new state because it is present in the input columns of the DFA but does not exist in the state column. Put “q0q1” in the state column along with its transitions.

States	0	1
→ q0	q0	q0q1
* q0q1	q0	q0q1



Problem 1



Construct a deterministic automaton equivalent to

$$M = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_0\})$$

where δ is defined by its state table (see Table 3.2).

TABLE 3.2 State Table for Example 3.6

State/ Σ	0	1
$\rightarrow \textcircled{q_0}$	q_0	q_1
q_1	q_1	q_0, q_1

For the deterministic automaton M_1 ,

- (i) the states are subsets of $\{q_0, q_1\}$, i.e. $\emptyset$, $[q_0]$, $[q_0, q_1]$, $[q_1]$;
- (ii) $[q_0]$ is the initial state;
- (iii) $[q_0]$ and $[q_0, q_1]$ are the final states as these are the only states containing q_0 ; and
- (iv) δ is defined by the state table given by Table 3.3.

TABLE 3.3 State Table of M_1 for Example 3.6

State/ Σ	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$[q_0]$	$[q_0]$	$[q_1]$
$[q_1]$	$[q_1]$	$[q_0, q_1]$
$[q_0, q_1]$	$[q_0, q_1]$	$[q_0, q_1]$

Problem 2



Find a deterministic acceptor equivalent to

$$M = (\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_2\})$$

where δ is as given by Table 3.4.

TABLE 3.4 State Table for Example 3.7

State/ Σ	a	b
$\rightarrow q_0$	q_0, q_1	q_2
q_1	q_0	q_1
$\odot q_2$		q_0, q_1

Solution

The deterministic automaton M_1 equivalent to M is defined as follows:

$$M_1 = (2^Q, \{a, b\}, \delta, [q_0], F')$$

where

$$F = \{[q_2], [q_0, q_2], [q_1, q_2], [q_0, q_1, q_2]\}$$

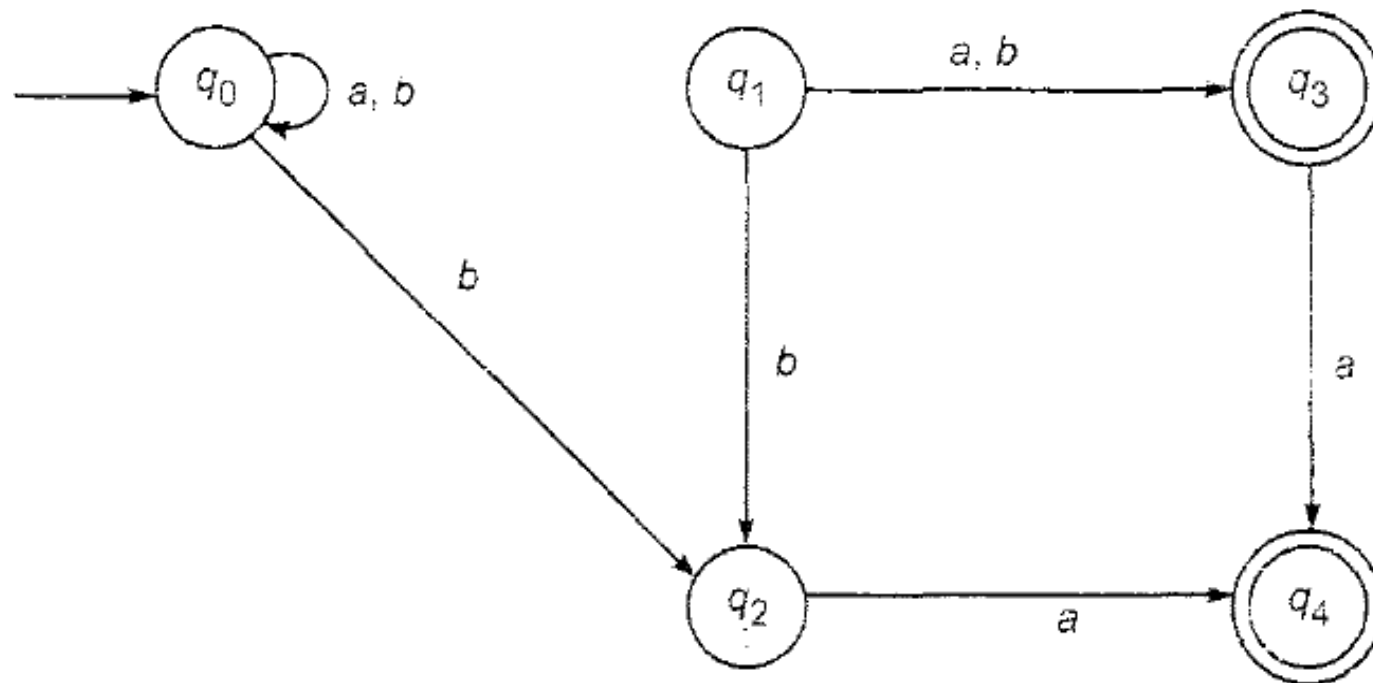
We start the construction by considering $[q_0]$ first. We get $[q_2]$ and $[q_0, q_1]$. Then we construct δ for $[q_2]$ and $[q_0, q_1]$. $[q_1, q_2]$ is a new state appearing under the input columns. After constructing δ for $[q_1, q_2]$, we do not get any new states and so we terminate the construction of δ . The state table is given by Table 3.5.

TABLE 3.5 State Table of M_1 for Example 3.7

State/ Σ	a	b
$[q_0]$	$[q_0, q_1]$	$[q_2]$
$[q_2]$	$\emptyset$	$[q_0, q_1]$
$[q_0, q_1]$	$[q_0, q_1]$	$[q_1, q_2]$
$[q_1, q_2]$	$[q_0]$	$[q_0, q_1]$

Practice example

Construct a DFA equivalent to the NFA M whose transition diagram is given



Two states q_1 and q_2 are equivalent (denoted by $q_1 \equiv q_2$) if

both $\delta(q_1, x)$ and $\delta(q_2, x)$ are final states, or both of them are nonfinal states for all $x \in \Sigma^*$.

As it is difficult to construct $\delta(q_1, x)$ and $\delta(q_2, x)$ for all $x \in \Sigma^*$ (there are an infinite number of strings in Σ^*), we give one more definition.

Two states q_1 and q_2 are k -equivalent ($k \geq 0$) if both $\delta(q_1, x)$ and $\delta(q_2, x)$ are final states or both nonfinal states for all strings x of length k or less. In particular, any two final states are 0-equivalent and any two nonfinal states are also 0-equivalent.

Following are the steps to minimize a DFA.

Step 1: Detect the unreachable states.

Step 2: Eliminate unreachable states (if any)

Step 3: Identify equivalent states and merge them.

3.1. Divide all states into two groups

Group A- Accepting states of automata.

Group B- Rejecting states of automata.

Repeat

For every group

For every state in the group

Find which group the input lead to if there are differences then partition the group into sets containing states which go to same groups under the inputs.

Until no more partition required.

3.2 Resulting final partition contains equivalent states now merge them into single state.

Step 4: Detect dead states.

Step 5: Eliminate dead states (if any).

Example 1:

Minimize the following DFA to reduced DFA:

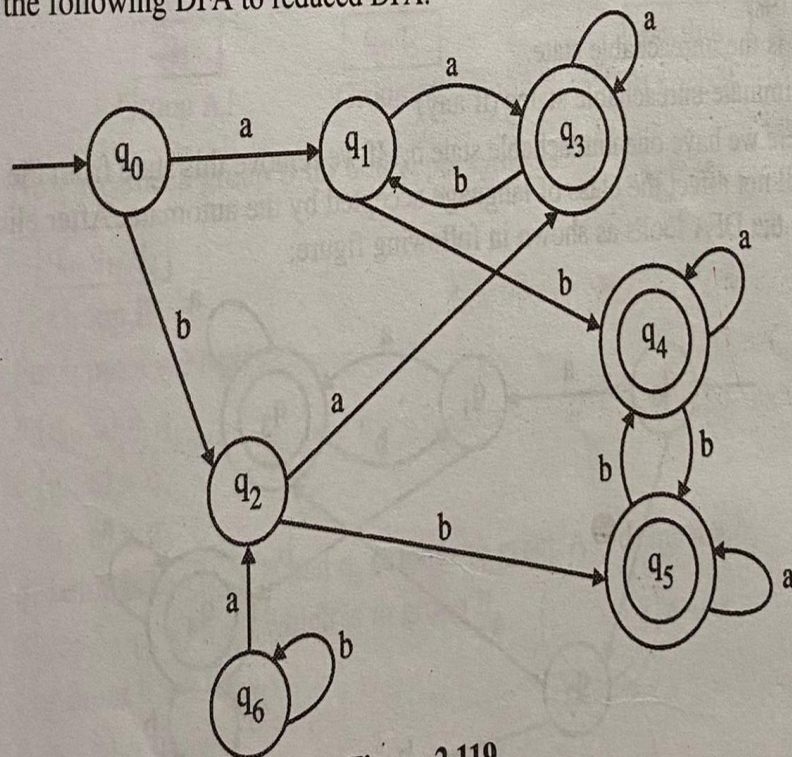


Figure 2.119

Now we have $T = \{q_0, q_1, q_2, q_3, q_4\}$

Again

$\delta(q_2, a) = q_3, \delta(q_2, b) = q_5$

Now we have $T = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

Again

$\delta(q_3, a) = q_3, \delta(q_3, b) = q_1$

No change in T because q_1, q_3 are already in set T .

Again

$\delta(q_4, a) = q_4, \delta(q_4, b) = q_5$

No change in T .

Again

$\delta(q_5, a) = q_5, \delta(q_5, b) = q_4$

Finally we have:

$T = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

Hence unreachable states are

$U = Q - T$

$= \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\} - \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$= \{q_6\}$

q_6 is the unreachable state.

Solution:

Step 1: Detect unreachable states.

i) $T = \{q_0\}$

ii) $\delta(q_0, a) = q_1, \delta(q_0, b) = q_2$

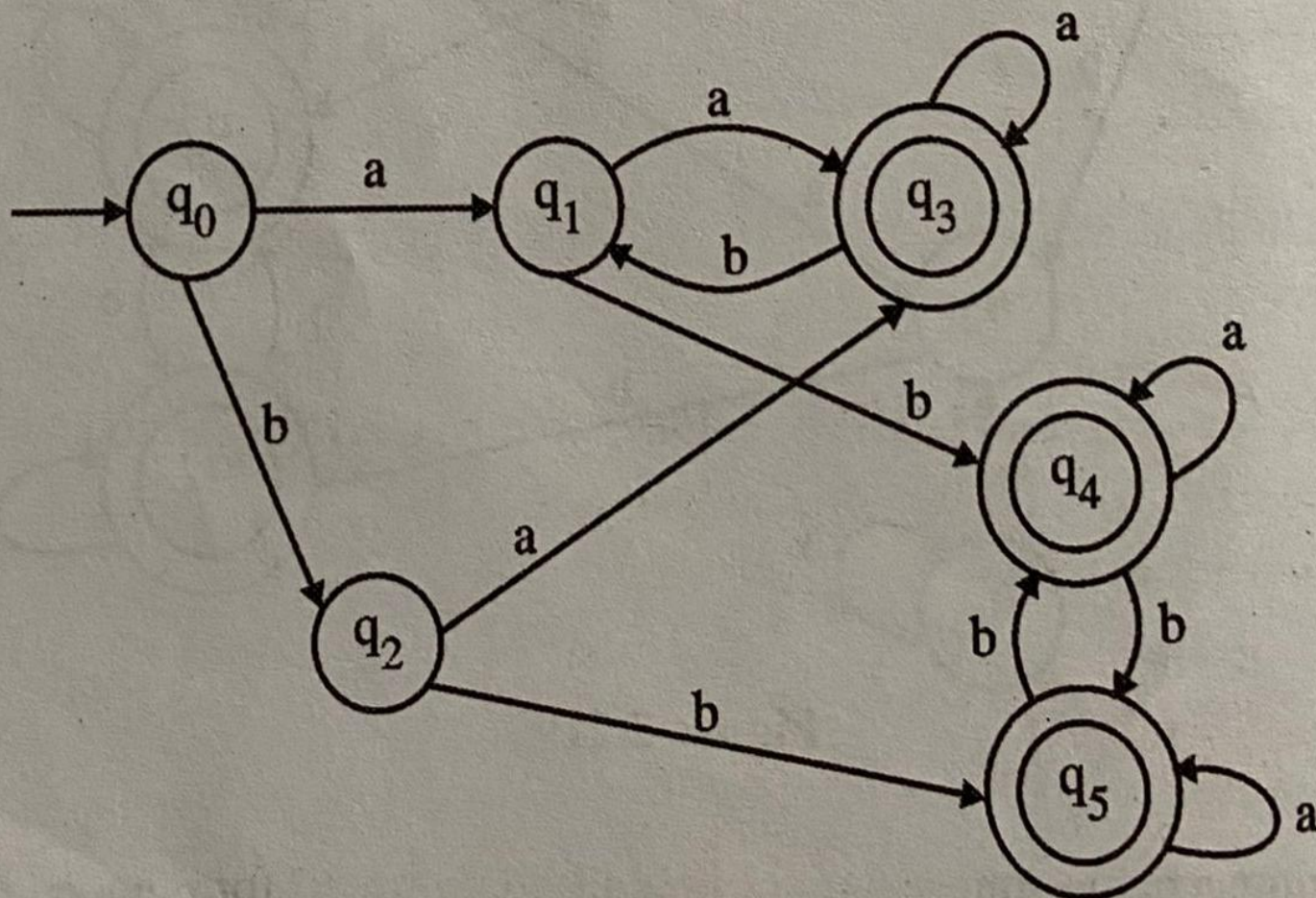
Now we have $T = \{q_0, q_1, q_2\}$

Again

$\delta(q_1, a) = q_3, \delta(q_1, b) = q_4$

Step 2: Eliminate unreachable states (if any)

Here we have one unreachable state q_6 . If we remove this state from the automata, it will not affect the class of language accepted by the automata. After eliminating q_6 , the DFA looks as shown in following figure:



Step 3: Identify equivalent states and merge them

To find equivalent states, first we divide all states into two groups.

$$\text{Group A} = \boxed{q_3, q_4, q_5}$$

$$\text{Group B} = \boxed{q_0, q_1, q_2}$$

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Group A contains all accepting states.

Group B contains all non accepting states.

Each state of every group find which group the input leads to. Now we will check both groups for both the input symbols.

First we check group A for input a.

$$\delta(q_3, a) = q_3$$

$$\delta(q_4, a) = q_4$$

$$\delta(q_5, a) = q_5$$

q_3, q_4, q_5 belongs to group A for input a.

Since all transitions for input a are in same group(group A). Now check group A for input b.

$$\delta(q_3, b) = q_1$$

$$\delta(q_4, b) = q_5$$

$$\delta(q_5, b) = q_4$$

q_1 belongs to group B for input b and q_4 and q_5 belongs to group A for input b.

Since $\delta(q_3, b) = q_1$, which is in group B. So we partition group A as:

$$\boxed{q_3}$$

Group A1

$$\boxed{q_4, q_5}$$

Group A2

$$\boxed{q_0, q_1, q_2}$$

Group B

q_3

Group A1

 q_4, q_5

Group A2

 q_0, q_1, q_2

Group B

Now we check group B for both input symbols.

 q_0, q_1, q_2

Group B

For input a we have:

$$\delta(q_0, a) = q_1$$

$$\delta(q_1, a) = q_3$$

$$\delta(q_2, a) = q_3$$

q_1 belongs to group B and q_3 belongs to group A1 for input a.

Since $\delta(q_0, a) = q_1$, which is in group B.

For input b we have:

$$\delta(q_0, b) = q_2$$

$$\delta(q_1, b) = q_4$$

$$\delta(q_2, b) = q_5$$

q_2 belongs to group B and q_4, q_5 belongs to group A2 for input b. So we partition group B as:

 q_0

Group B1

 q_1, q_2

Group B2

After partitions group A and group B. We have following groups:

 q_3

Group A1

 q_4, q_5

Group A2

 q_0

Group B1

 q_1, q_2

Group B2

Since further division of group A1 and group B1 is not possible. For group A2 on input a we have:

$$\delta(q_4, a) = q_4$$

$$\delta(q_5, a) = q_5$$

q_4 and q_5 belongs to group A2 for input a. For input b we have:

$$\delta(q_4, b) = q_5$$

$$\delta(q_5, b) = q_4$$

q_4 and q_5 belongs to group A2 for input b.

Since both belongs to same group so further division is not possible for group A2.

For group B2 on input a, we have:

$$\delta(q_1, a) = q_3$$

$$\delta(q_2, a) = q_3$$

q_3 belongs to group A1 for input a.

For input b, we have

$$\delta(q_1, b) = q_4$$

$$\delta(q_2, b) = q_5$$

q_4 and q_5 belongs to group A2 for input b.

No further partitioning is possible. So we have finally following groups.

 q_3

Group A1

 q_4, q_5

Group A2

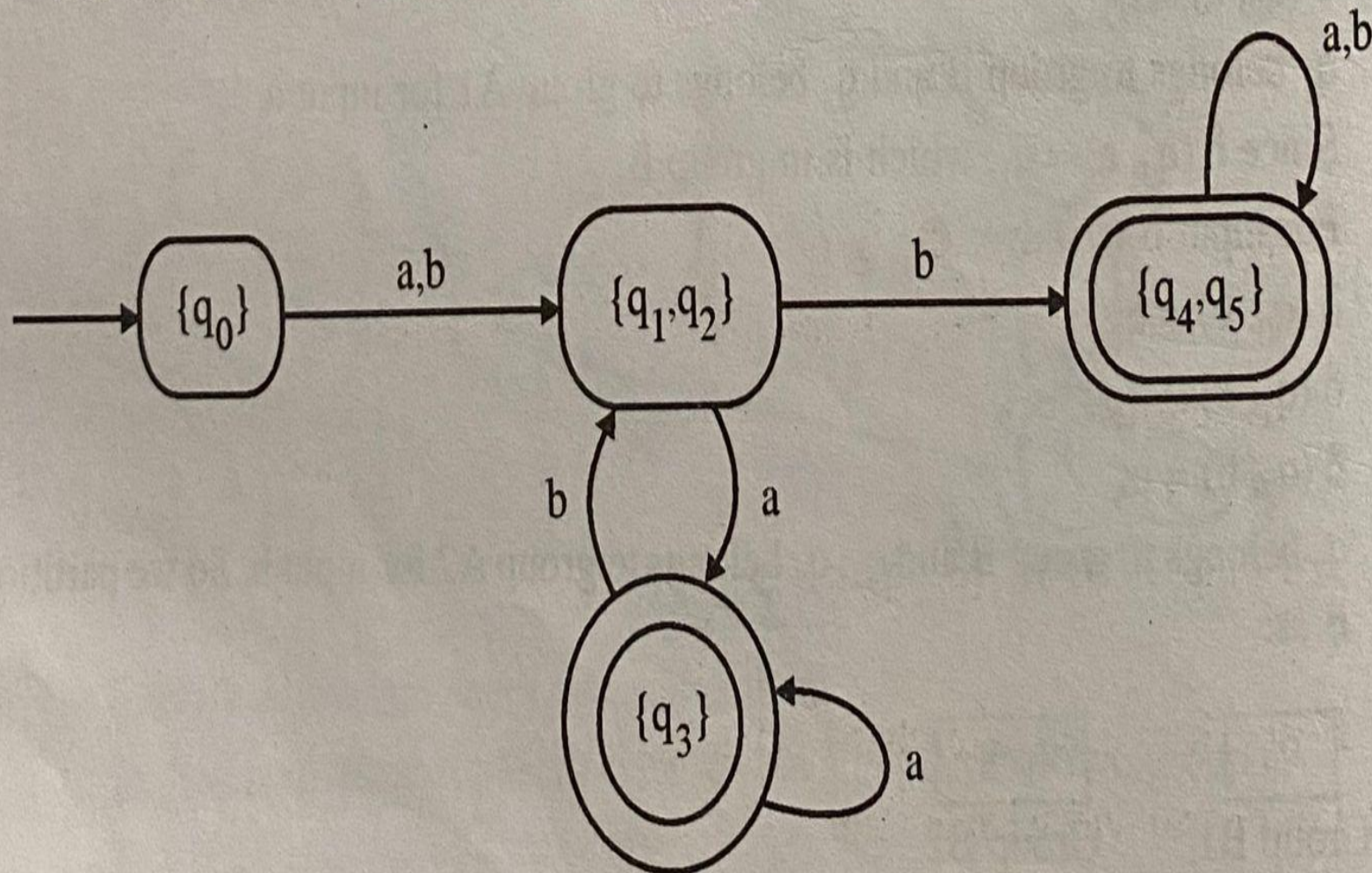
 q_0

Group B1

 q_1, q_2

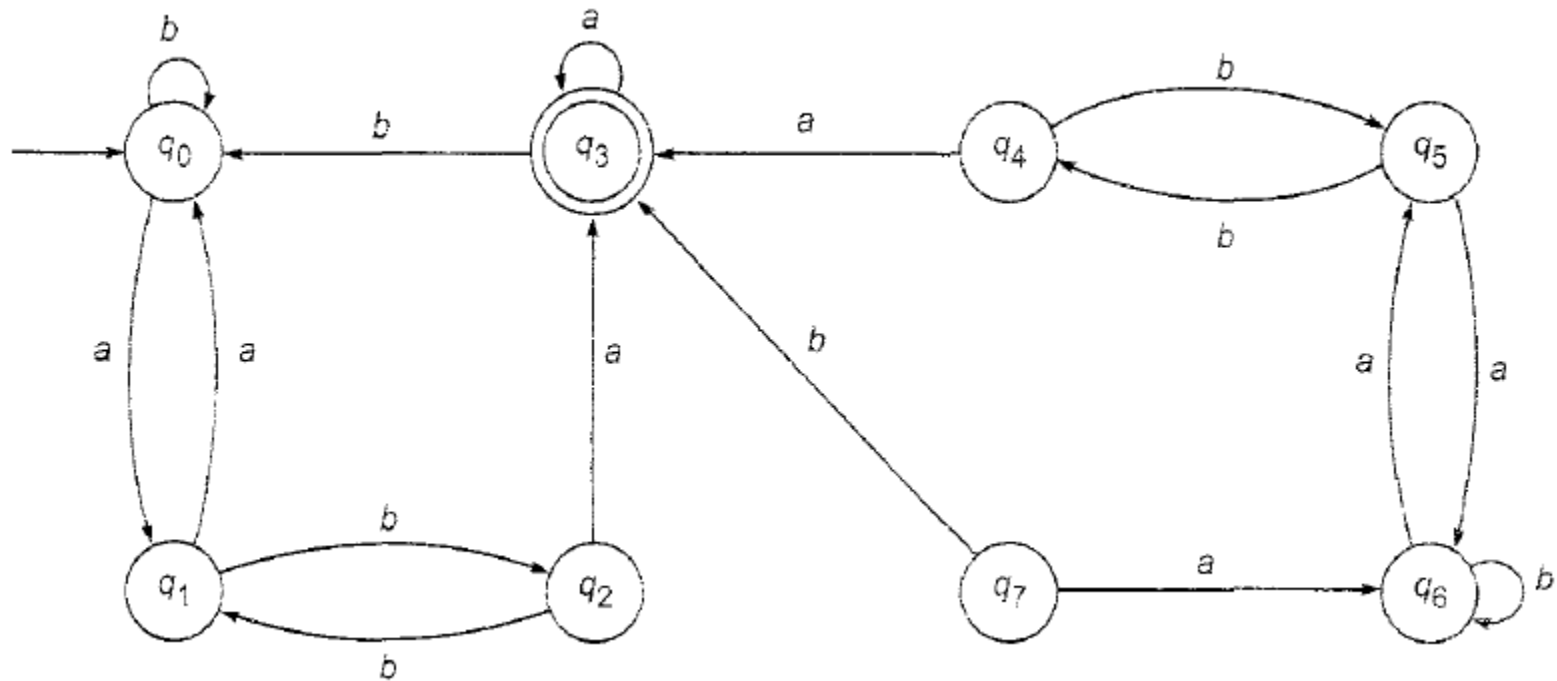
Group B2

We concluded that q_4, q_5 are equivalent and also q_1, q_2 are equivalent. So merge them and the resultant automata is given in following figure:



Practice Example

Construct the minimum state automaton equivalent to the transition diagram



Solution

